FORWARD/REVERSE HYDRAULIC PRESSURE TEST

1. Park the machine on a solid flat surface and stop the engine. Set the parking brake and block the front drum securely.
2. Check hydraulic oil level by viewing the hydraulic oil sight glass. The hydraulic oil level must be below the top and above the bottom of the sight glass. DO NOT OVERFILL!
3. Adjust engine RPM (3,500 +/- 50 RPM).
4. Let engine run for 5 minutes, this will bring the hydraulic oil operating temperature to a minimum of 135° F.
5. Check and repair all hydraulic leaks.
6. On the manifold block, install a 5,000 psi pressure gauge (Figure 29) to the forward pressure quick disconnect test port 2.
7. Run engine at full throttle.
8. Move the travel lever to the forward position. MAKE SURE FRONT AND REAR DRUMS DO NOT SPIN.
9. Read the pressure gauge. The relief pressure (roller drums blocked) will read 2,900 ± 145 psi. Under normal operating conditions this pressure will read 400-600 psi.
10. Return the travel lever to the neutral position and stop the engine.
11. Install the pressure gauge into the reverse quick disconnect port 3 and repeat the above procedures. The relief pressure reading for the reverse test port 3 will be the same (2,900 psi). Again normal operation pressure for the reverse port will be 400-600 psi.
12. Normal operating pressures are based on the machine traveling on level, firm surface. The operating pressures will increase significantly when traveling uphill.

VIBRATION CIRCUIT PRESSURE TEST

1. Place the front drum on soil, gravel, or a heavy rubber mat. DO NOT ACTIVATE VIBRATION FEATURE ON CONCRETE OR HARD SURFACE!
2. On the manifold block install a 5,000 psi pressure gauge to quick disconnect test port 1.
3. Start the engine and run at full throttle.
4. To start the vibration, press the pushbutton switch (vibration control) located on top of the travel lever. Under normal operating conditions this pressure will range between 900-1,500 psi.

NOTICE

Vibration relief pressure is hard to read accurately using this test. It may be necessary to disconnect the pressure line to the vibration drive motor. Plug this line and retest, pressure will be exact relief pressure. DO NOT perform this test for a long period of time — damage could occur.

STEERING PRESSURE TEST

1. On the manifold block, install a 5,000 psi pressure gauge into quick disconnect test port 1 (Figure 29).
2. Start the engine and run at full throttle.
3. Turn the steering wheel to the left or right (maximum) and hold. Read the steering relief pressure. The relief pressure reading for the steering test port 1 will be 700 psi. Again, under normal operating conditions, pressure for the steering port will be between 200-400 psi.
4. If maximum pressure cannot be reached, plug the steering cylinder pressure hoses and retest. If pressure is correct, the steering cylinder is leaking. DO NOT ACTIVATE THE VIBRATION FUNCTION WHILE PERFORMING THIS TEST!

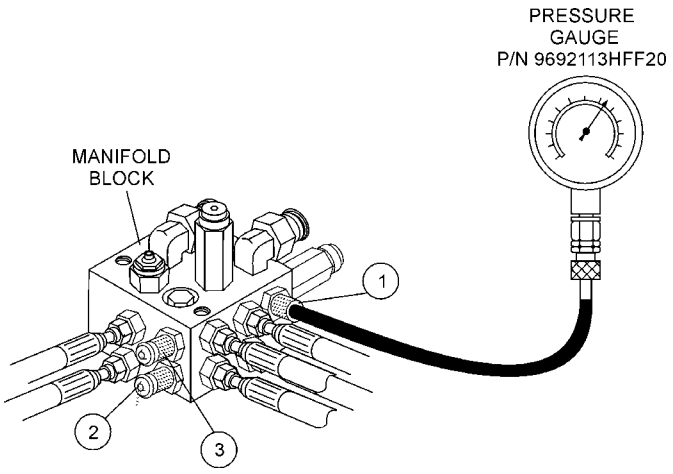


Figure 29. Manifold Test Ports

FORWARD/REVERSE RELIEF VALVES DRIVE ADJUSTMENT

The forward and reverse relief valves (Figure 30) are located in the hydrostatic pump (not on the manifold block) under plugs 1 and 2 . Plug 2 is located in the same location as plug 1, except it is on the bottom side of the pump. Relief valve 1 is reverse and 2 is forward.

NOTICE

Pumps supplied after 01/01/2003 do not have adjustable relief cartridges. The pressure is fixed at 2,900 ± 145 psi. If the relief pressure is not within normal range, the complete cartridge must be replaced.

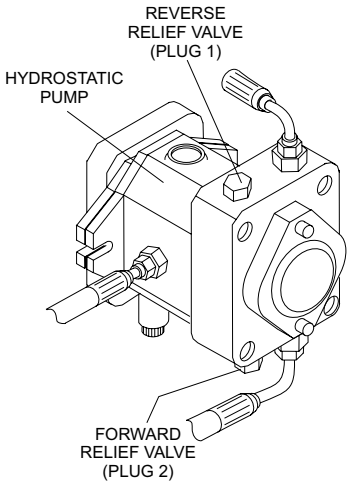


Figure 30. Adjusting Relief Valves

1. Clean the area around the cap.
2. Remove the cap.
3. Carefully remove the valve cartridge.
4. Using an allen wrench, remove the top nut (extreme care must be taken not to loosen existing shims, spring, or valve) Add or remove shims as required. Adding shims increases pressure; removing shims lowers pressure. One shim is equal to approximately 50 psi. Shim P/N is 34538.
5. Install the top nut and install valve in the pump cavity using extreme care not to bind.
6. Install the valve cap.
7. Retest pressure, further adjustment may be needed if pressure is not correct.

STEERING RELIEF VALVE PRESSURE ADJUSTMENT

To adjust the relief valve steering pressure perform the following:

1. Insert an 8 mm allen wrench into steering relief port 4 (Figure 36) on the manifold block.

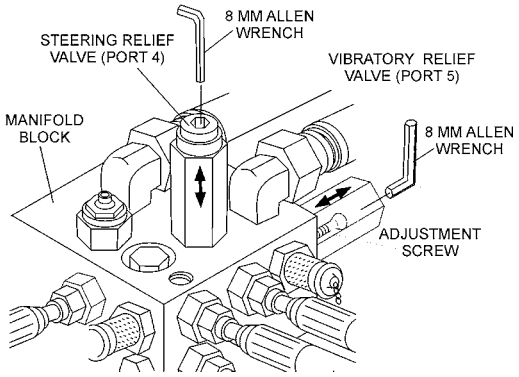


Figure 31. Steering and Vibration Relief Valves

2. On the manifold, connect at 5,000 psi pressure gauge into quick disconnect test port 1.
3. Start the engine and run at full throttle.
4. Turn the steering wheel to the left or right (maximum) and hold. Read the steering relief pressure. The relief pressure reading for the steering test port 1 will should be 700 psi.
5. If the steering relief pressure is not 700 psi, using the allen wrench adjust the pressure at port 4 until the pressure gauge reads 500 psi.

VIBRATION RELIEF VALVE PRESSURE ADJUSTMENT

1. Insert an 8 mm allen wrench into steering relief port 5 (Figure 31) on the manifold block.
2. On the manifold block, insert a 5,000 psi pressure gauge into quick disconnect test port 1.
3. Start the engine and run at full throttle.
4. To start the vibration, press the pushbutton switch (vibration control) located on top of the travel lever. The relief pressure should read 900 ~ 1,500 psi.
5. If the vibration relief pressure is not 900 ~ 1,500 psi, using the allen wrench, adjust the pressure at port 5 until the pressure gauge reads correctly.

REMOVING AND REPLACING HYDROSTATIC PUMP

1. Set the parking brake.
2. Disconnect the battery.
3. Clean the pump and all connections.
4. Mark and disconnect all hoses and lines from the pump.
5. Disconnect the forward / reverse control cable.
6. Disconnect the pump support bracket.
7. Remove the engine mounting bolts.
8. Elevate the pump and engine assembly using a proper lifting device.
9. Disconnect and remove the hydrostatic pump assembly.
10. Repair or replace the hydrostatic pump as required.
11. Install the hydrostatic pump in the reverse order of removal, using Locktite 271 on all mounting bolts and nuts.
12. Test operation. Test and adjust the forward and reverse relief pressures as required. Adjust the forward/reverse control cable.

REMOVING AND REPLACING VIBRATION/ STEERING PUMP

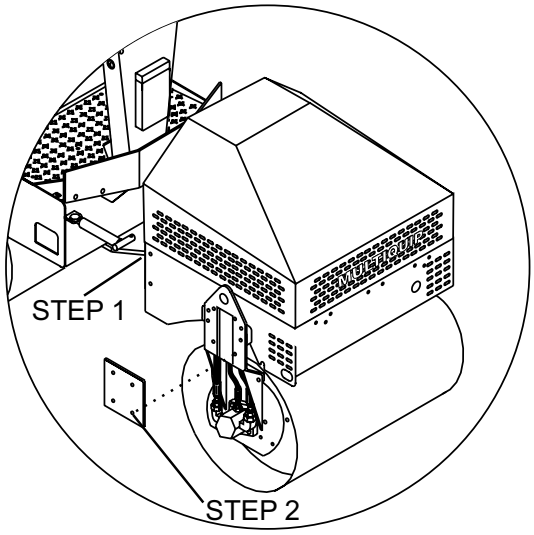
1. Remove the hydrostatic pump as per preceding instructions.
2. Remove all hoses and lines.
3. Disconnect the vibration / steering pump and remove.
4. Repair or replace pump as required.
5. Install the pump in the reverse order of removal, using Locktite 271 on all mounting bolts and nuts.
6. Test operate. Test and adjust the forward and reverse pressure relief valves as required. Adjust the forward/ reverse control cable. Test and adjust the vibration and steering pressure relief valves as required.

DRUMS AND MAIN FRAME

1. The front drum is designed to apply vibration and compaction force to the operating surface for compaction. This vibration and compaction force is produced when the vibrator shaft is rotated. Maximum efficiency is achieved only when the engine is operated at full throttle.
2. A single drive motor is mounted on the left side of the drum and is shock mounted. This type of drive motor is designed for maximum torque and power.
3. The vibrator is driven by a gear motor coupled to the vibrator shaft. The vibrator assembly rotates inside of a sealed housing containing oil to lubricate the bearings. This side of the drum is also shock mounted.

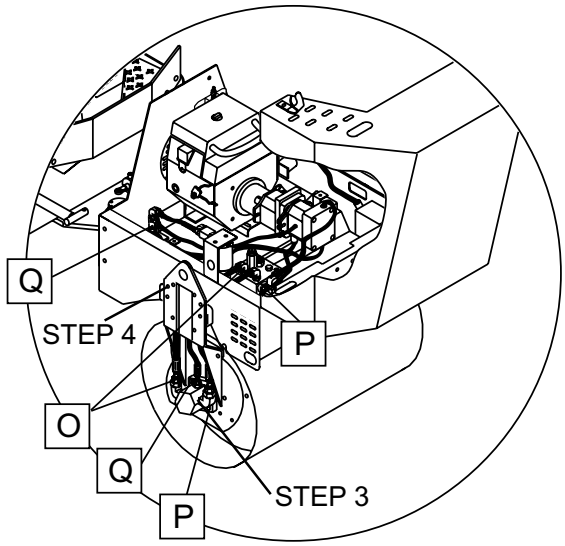
FRONT AND REAR DRUM REMOVAL AND DISASSEMBLY.

Refer to Figure 32, Figure 33, and Figure 34 for the removal and disassembly of the front and rear drums. When reassembling the front drum, use Locktite 271 on all mounting hardware.



1. JACK FRAME UP SLIGHTLY AT ARTICULATION JOINT.

2. REMOVE THE FOUR ALLEN HEAD SCREWS HOLDING THE SIDE COVER OF THE DRUM SUPPORT PLATE.

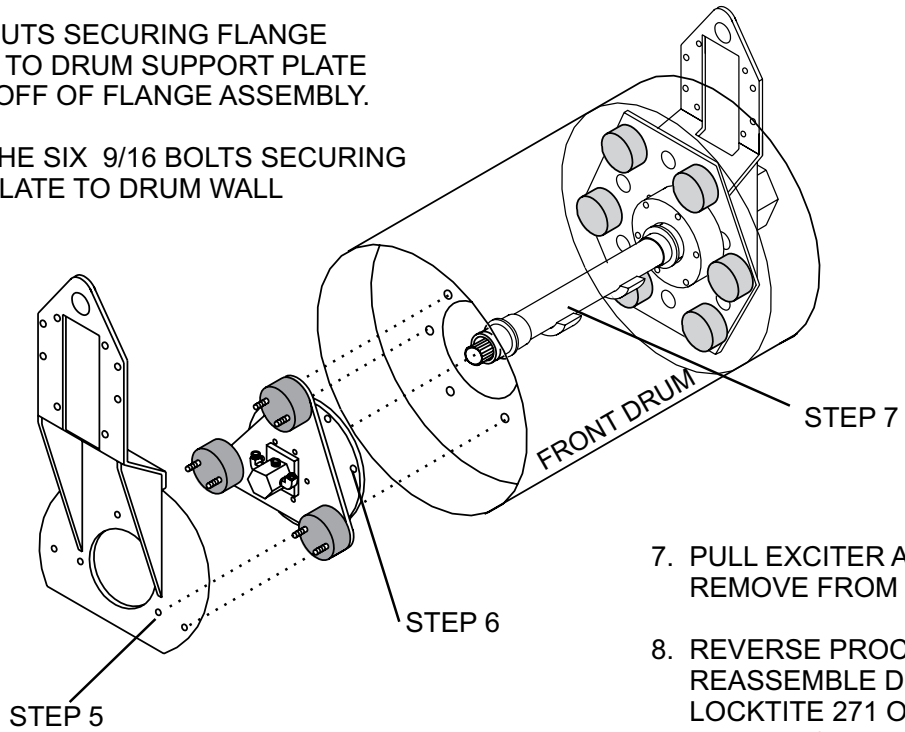


3. MARK AND DISCONNECT HOSES FROM MOTOR AND MANIFOLD.

4. REMOVE FOUR ALLEN HEAD SCREWS FROM DRUM SUPPORT PLATES.(SCREWS MAY NEED TO BE HEATED TO REMOVE)

5. REMOVE NUTS SECURING FLANGE ASSEMBLY TO DRUM SUPPORT PLATE AND PULL OFF OF FLANGE ASSEMBLY.

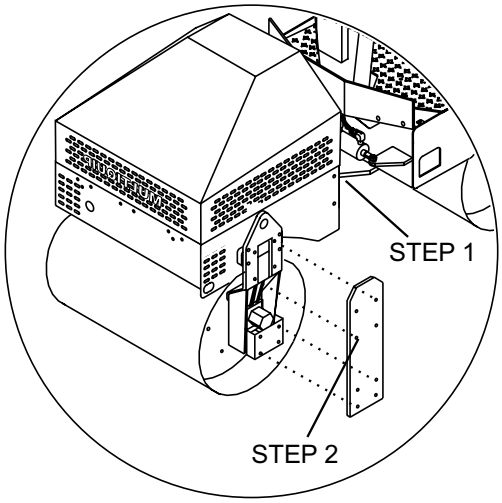
6. REMOVE THE SIX 9/16 BOLTS SECURING EXCITER PLATE TO DRUM WALL

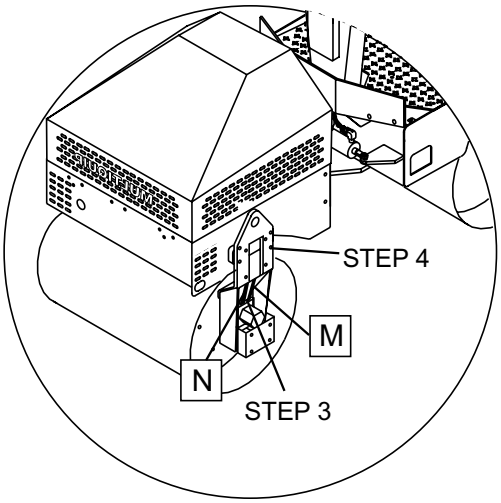


7. PULL EXCITER ASSEMBLY TO REMOVE FROM DRUM INTERIOR.

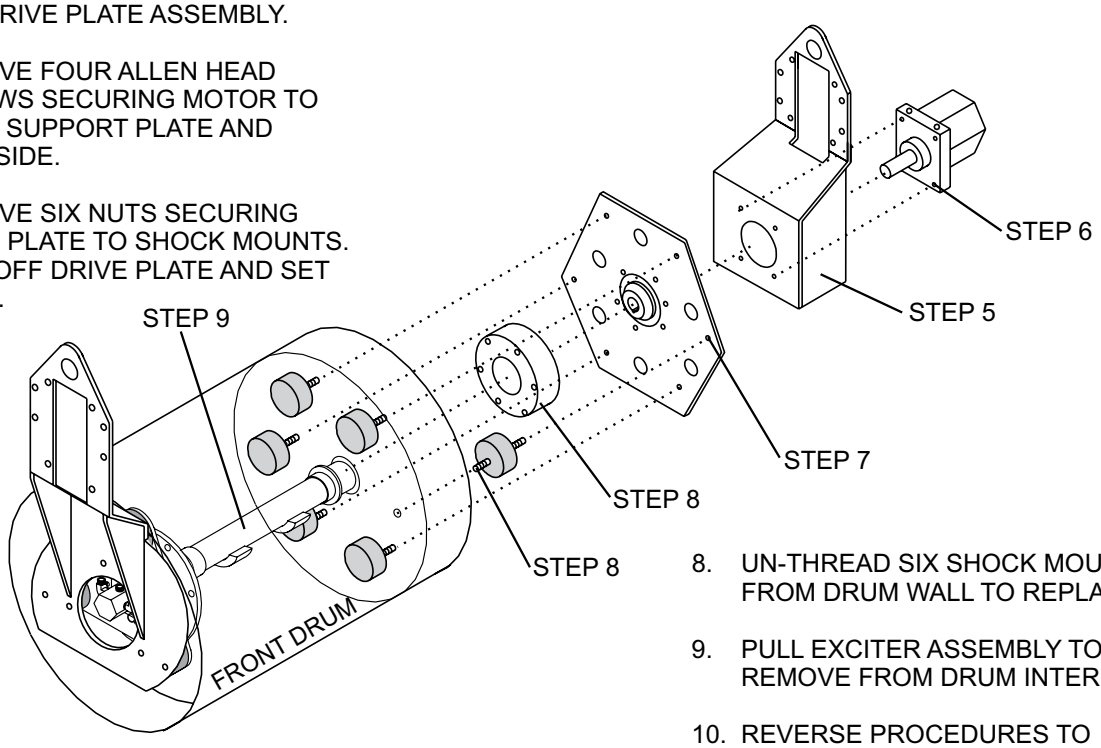
8. REVERSE PROCEDURES TO REASSEMBLE DRUM. USE LOCKTITE 271 ON ALL BOLTS AND NUTS.

Figure 32. Front Right Drum Removal



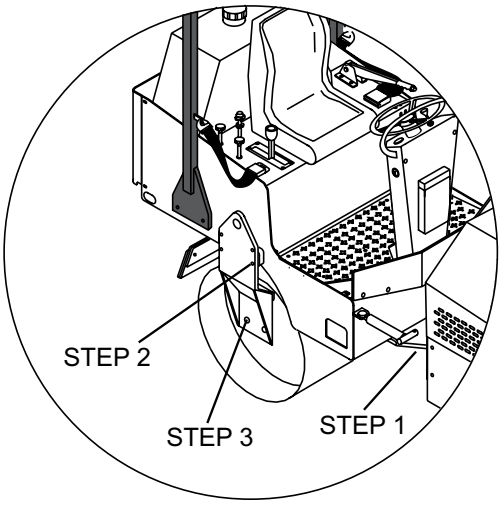


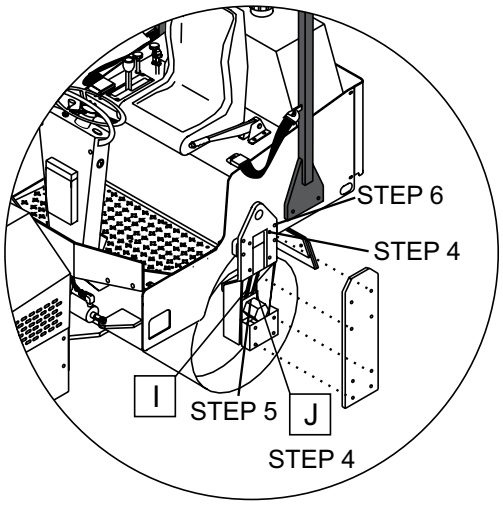
1. JACK FRAME UP SLIGHTLY AT ARTICULATION JOINT.
2. REMOVE THE EIGHT ALLEN HEAD SCREWS HOLDING THE SIDE COVER OF THE DRUM SUPPORT PLATE.
3. MARK AND DISCONNECT HOSES FROM MOTOR.
4. REMOVE FOUR ALLEN HEAD SCREWS FROM DRUM SUPPORT PLATE. (SCREWS MAY NEED TO BE HEATED TO REMOVE)
5. PULL DRUM SUPPORT PLATE OFF DRIVE PLATE ASSEMBLY.
6. REMOVE FOUR ALLEN HEAD SCREWS SECURING MOTOR TO DRUM SUPPORT PLATE AND SET ASIDE.
7. REMOVE SIX NUTS SECURING DRIVE PLATE TO SHOCK MOUNTS. PULL OFF DRIVE PLATE AND SET ASIDE.



8. UN-THREAD SIX SHOCK MOUNTS FROM DRUM WALL TO REPLACE.
9. PULL EXCITER ASSEMBLY TO REMOVE FROM DRUM INTERIOR.
10. REVERSE PROCEDURES TO REASSEMBLE DRUM. USE LOCKTITE ON ALL NUTS AND BOLTS.

Figure 33. Front Left Drum Removal





1. JACK FRAME UP SLIGHTLY AT ARTICULATION JOINT.

2. REMOVE THE FOUR ALLEN HEAD SCREWS SECURING THE DRUM SUPPORT PLATE TO THE FRAME.

3. REMOVE NUTS SECURING BEARING FLANGE TO BASE OF DRUM SUPPORT PLATE.

4. REMOVE ALLEN HEAD SCREWS HOLDING THE SIDE COVER OF THE DRUM SUPPORT PLATE.

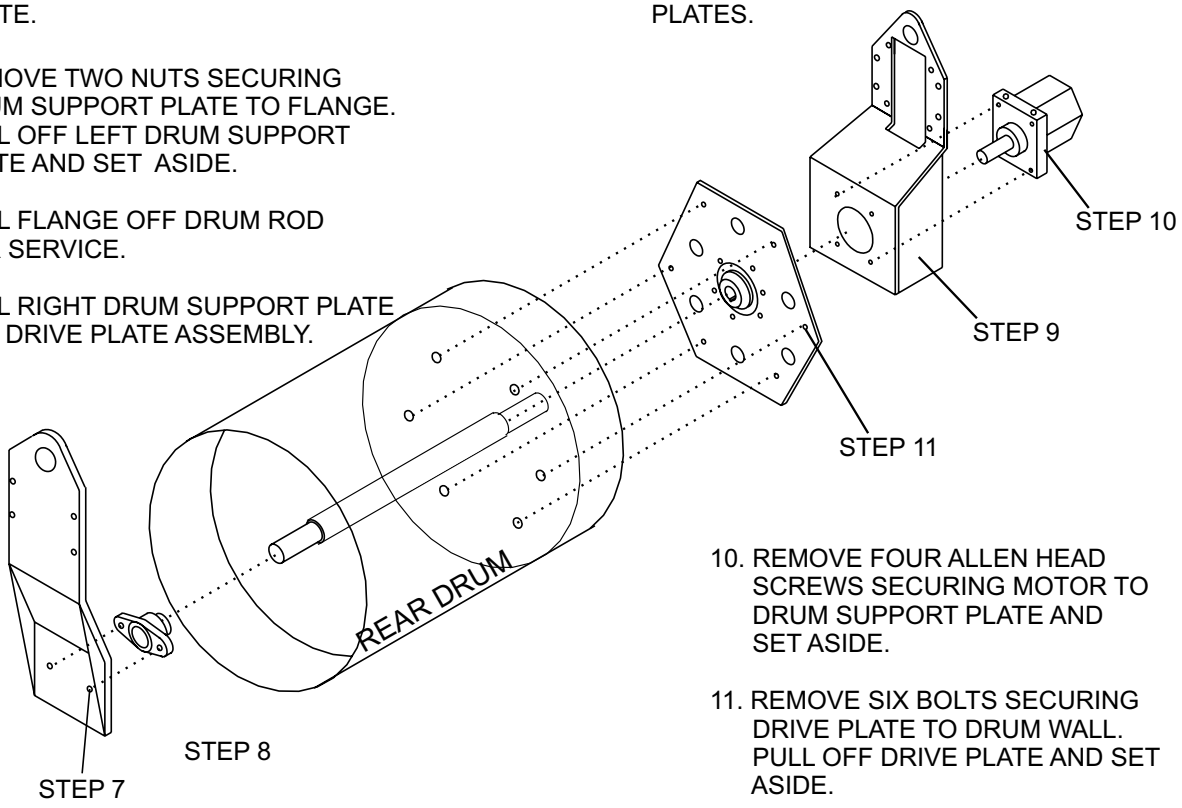
5. MARK AND DISCONNECT HOSES FROM MOTOR.

6. REMOVE FOUR ALLEN HEAD SCREWS FROM DRUM SUPPORT PLATES.

7. REMOVE TWO NUTS SECURING DRUM SUPPORT PLATE TO FLANGE. PULL OFF LEFT DRUM SUPPORT PLATE AND SET ASIDE.

8. PULL FLANGE OFF DRUM ROD FOR SERVICE.

9. PULL RIGHT DRUM SUPPORT PLATE OFF DRIVE PLATE ASSEMBLY.



10. REMOVE FOUR ALLEN HEAD SCREWS SECURING MOTOR TO DRUM SUPPORT PLATE AND SET ASIDE.

11. REMOVE SIX BOLTS SECURING DRIVE PLATE TO DRUM WALL. PULL OFF DRIVE PLATE AND SET ASIDE.

12. REVERSE PROCEDURES TO REASSEMBLE DRUM.

Figure 34. Rear Drum Removal/Installation

ALTERNATOR/REGULATOR

This roller is equipped with a 20-amp charging system. This system uses three charge coils connected in parallel. The windings of each coil are wound with heavy-duty insulated wire and are further protected by insulating material for long life. A voltage regulator is provided to control the amount of charge voltage being delivered to the 12 volt electrical system. For servicing this charging system, contact your Honda dealer.

NEUTRAL SAFETY SWITCH

The travel lever is provided with a neutral safety switch (Figure 35) that prevents the engine from starting when the lever is in the forward or reverse drive position. Lever must be placed in the neutral position in order for the starter to function.

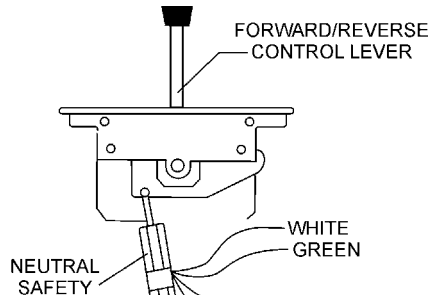


Figure 35. Safety Switch

To check this switch, remove the panel located below and to the rear of the floor. This switch is mounted to the lower forward/ reverse lever mechanism. Disconnect the white and green wires. Place the lever in neutral and test for continuity. Place the lever in forward or reverse; no continuity should be present. Repair or replace as required.

VIBRATOR SOLENOID

The vibrator solenoid (Figure 36) is located in the manifold valve block and is controlled by the vibration switch located on top of the travel lever. This solenoid controls the vibration control valve which supplies hydraulic oil to the vibration drive motor.

To test the vibration solenoid, turn the ignition switch on. Check for proper supply voltage and ground. If proper voltage and ground are present, check the solenoid for continuity. If no continuity, replace the solenoid coil. If continuity is present, place your hand on the solenoid and turn the switch to the ON position; movement should be felt. DO NOT START THE ENGINE FOR THIS TEST.

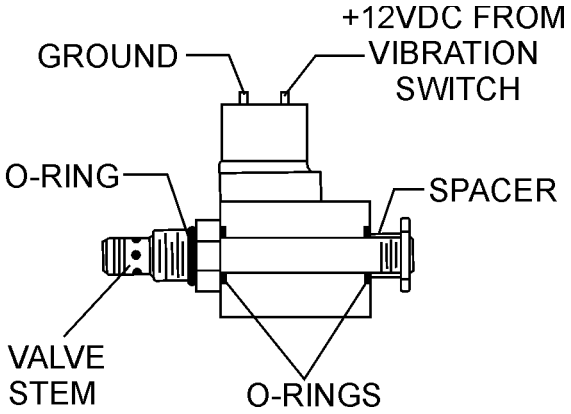


Figure 36. Vibration Solenoid

BATTERY

Mishandling of the battery shortens the service life of the battery and adds to maintenance cost. When handling the battery do the following:

- Be careful not to let the battery electrolyte come in contact with your body or clothing.
- Always wear eye protection and rubber gloves, since the battery contains sulfuric acid which burns skin and eats through clothing.
- Check the battery regularly and make sure that each electrolyte level is to the bottom of the vent well (Figure 42). If necessary, add only distilled water in a well-ventilated area.

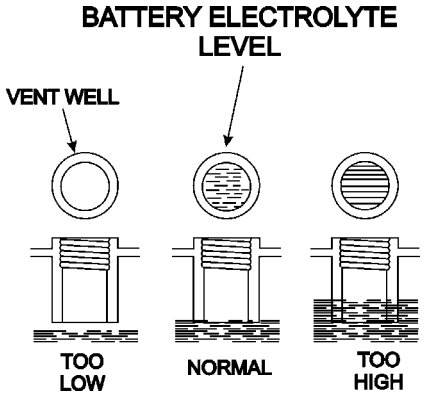
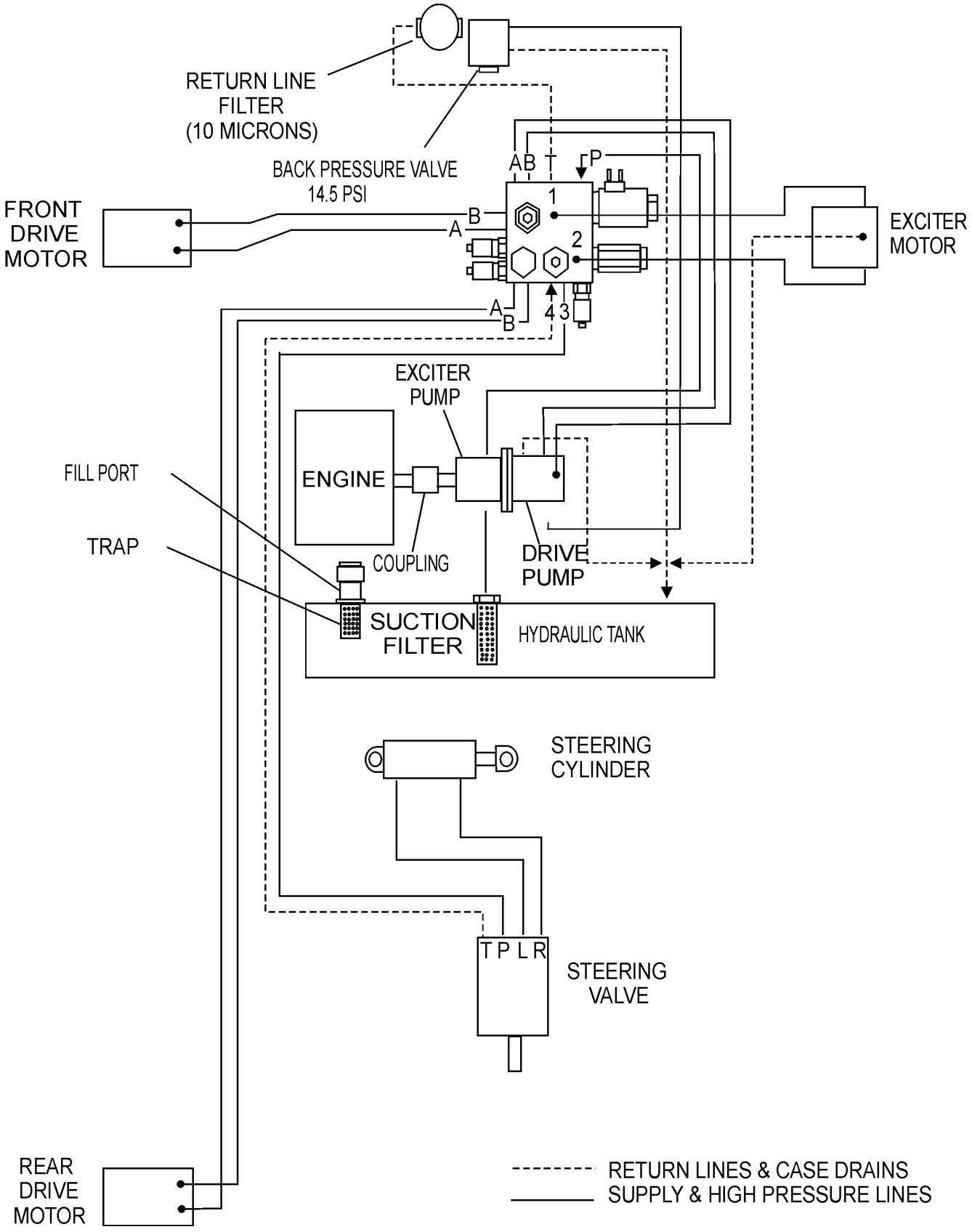


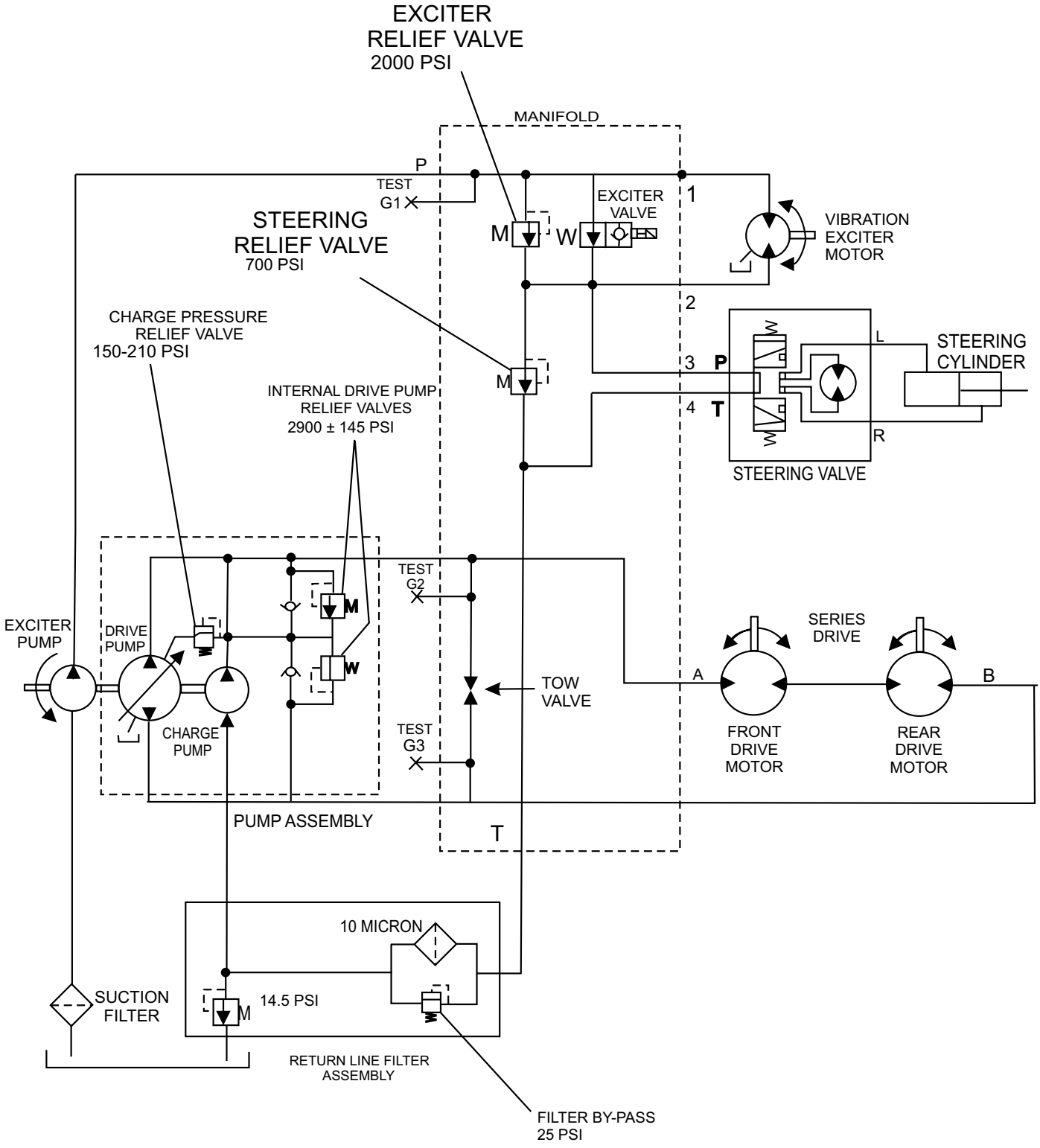
Figure 37. Battery Fluid Levels

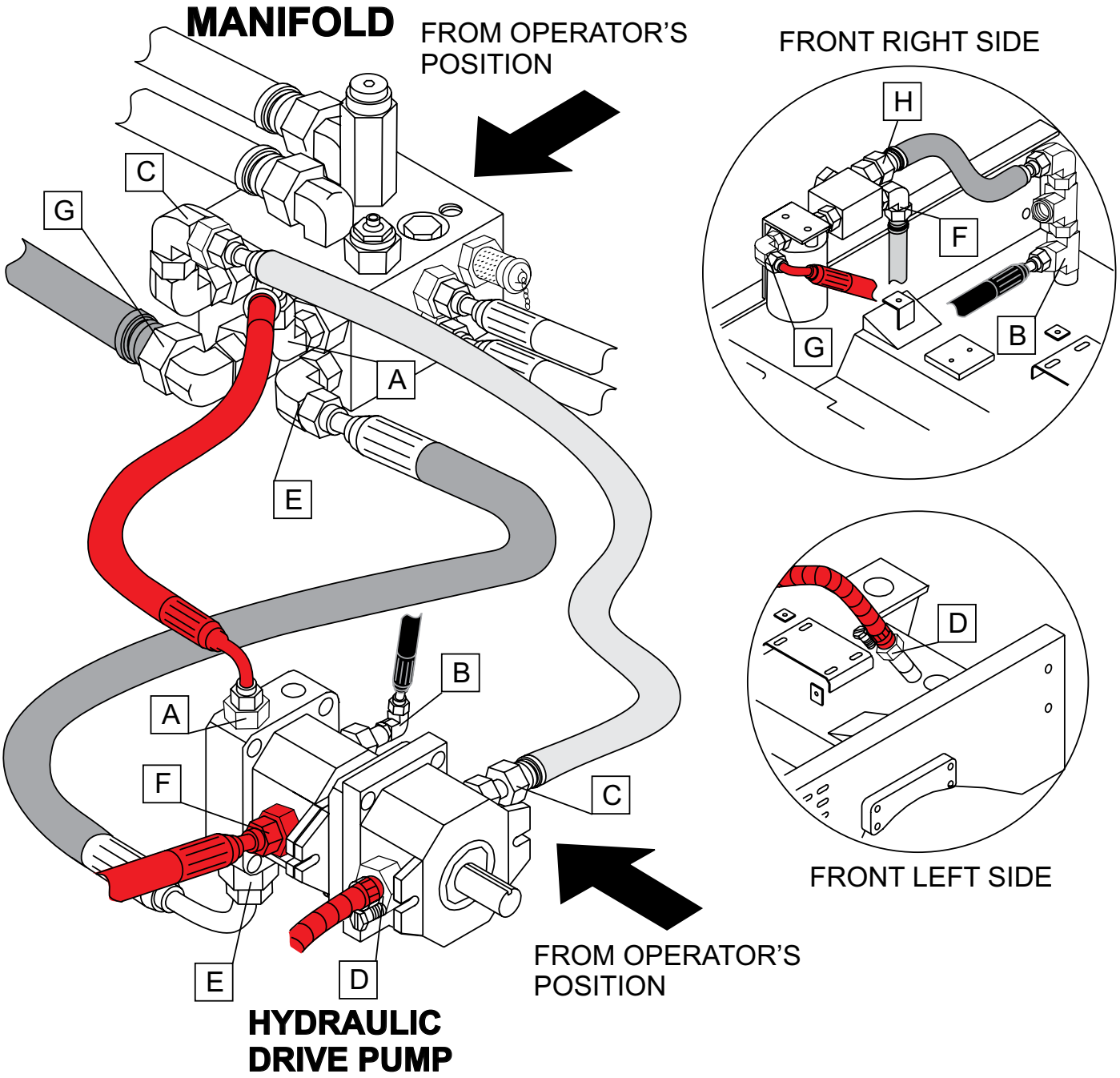
ROLLER STORAGE

For storage of the roller for over 30 days, the following is recommended:

1. Drain the fuel tank completely, or add STA-BIL to the fuel.
2. Run the engine until the fuel in the injection system is completely consumed.
3. Completely drain used oil from the engine crankcase and fill with fresh clean oil, then follow the procedures described in the engine manual for engine storage.
4. Drain water tank.
5. Clean the entire roller and engine compartment.
6. Remove battery and store it in cool dry place.
7. Cover the roller and place it a clean dry area, that is protected from harsh elements.
8. Remove ignition key, and store in a safe place.

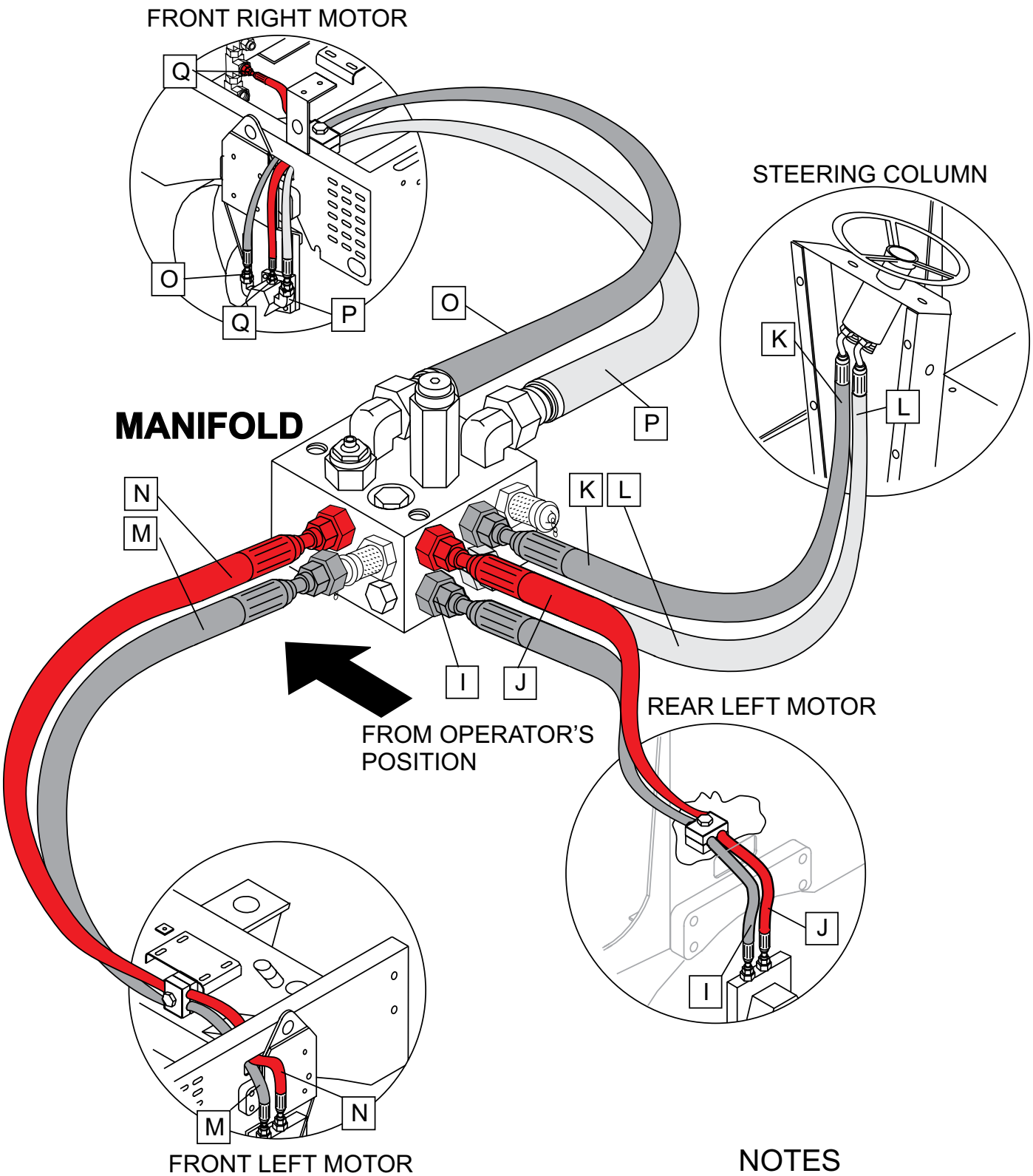

RETURN LINE
FILTER
(10 MICRONS)
BACK PRESSURE VALVE
14.5 PSI
AB T P
1
2
B
A
A
B
4 3
FRONT
DRIVE
MOTOR
EXCITER
MOTOR
EXCITER
PUMP
ENGINE
FILL PORT
TRAP
COUPLING
DRIVE
PUMP
SUCTION
FILTER
HYDRAULIC TANK
STEERING
CYLINDER
T P L R
STEERING
VALVE
REAR
DRIVE
MOTOR
RETURN LINES & CASE DRAINS
SUPPLY & HIGH PRESSURE LINES


EXCITER
RELIEF VALVE
2000 PSI
MANIFOLD
P
TEST
G1
STEERING
RELIEF VALVE
700 PSI
M
W
EXCITER
VALVE
1
VIBRATION
EXCITER
MOTOR
2
CHARGE PRESSURE
RELIEF VALVE
150-210 PSI
M
INTERNAL DRIVE PUMP
RELIEF VALVES
2900 ± 145 PSI
3
P
4
T
L
R
STEERING
CYLINDER
STEERING VALVE
TEST
G2
EXCITER
PUMP
DRIVE
PUMP
CHARGE
PUMP
TOW
VALVE
TEST
G3
A
SERIES
DRIVE
B
FRONT
DRIVE
MOTOR
REAR
DRIVE
MOTOR
PUMP ASSEMBLY
T
10 MICRON
SUCTION
FILTER
14.5 PSI
RETURN LINE FILTER
ASSEMBLY
FILTER BY-PASS
25 PSI



NOTES

HOSES SHADED FOR VISUAL CLARITY.

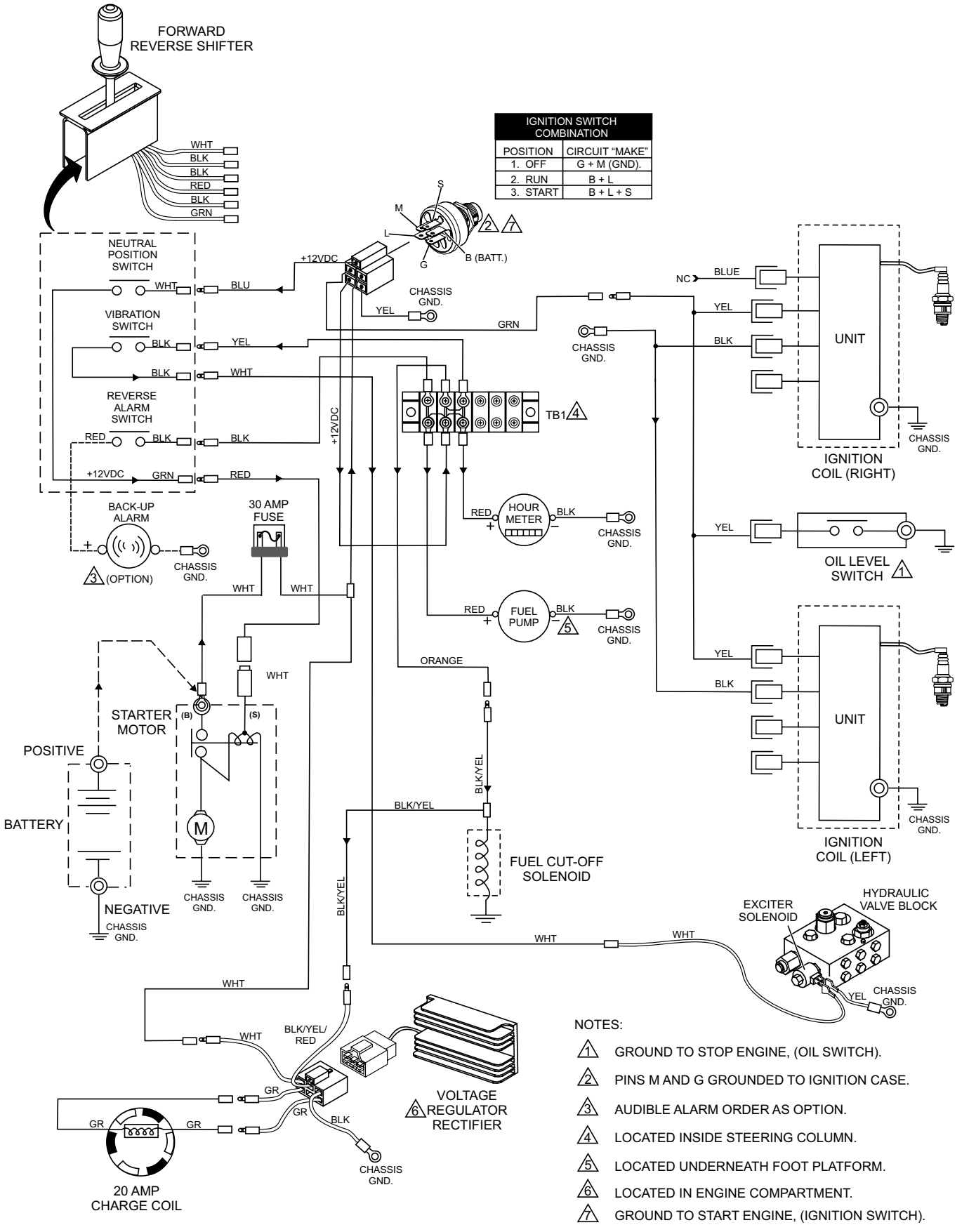


NOTES

HOSES SHADED FOR VISUAL CLARITY.

Troubleshooting (Roller)		
SYMPTOM	**POSSIBLE PROBLEM**	**SOLUTION**
Low Vibration	Low engine RPM?	Check and adjust engine speed.
	Low hydraulic pressure?	Check and adjust hydraulic pressure.
No Vibration	No voltage to 12 volt solenoid?	Check vibration switch, check coil for continuity.
	No oil pressure?	Check for proper oil pressure or defective relief valve.
	Broken motor coupler?	Replace coupler.
Slow Steering	Low hydraulic pressure?	Check pressure.
	Steering column connector?	Inspect bearings and grease if necessary.
	Defective steering cylinder?	Check seals, replace if necessary.
Steers Slow in One Direction	Defective steering cylinder?	Inspect cylinder for contamination, replace if necessary.
Slow Forward/Reverse Speed	Low hydraulic pressure?	Check pressure.
	Rubber scrapper adjustment incorrect?	Check for proper adjustment against drum.
	Rubber shock mount loose?	Inspect for broken mount, replace if necessary.
	Free wheel valve "open"?	Close valve.
	Speed control cable out of adjustment?	Inspect cable and adjust if necessary.
Main Hydraulic Pressure Low	Incorrect engine speed?	Adjust engine speed to correct speed.
	Broken rubber shock mounts or drive plate?	Inspect for cracks, replace if necessary.

Troubleshooting (Engine)		
Symptom	**Possible Problem**	**Solution**
Difficult to start, fuel is available, but no spark at spark plug.	Spark plug bridging?	Check gap, insulation or replace spark plug.
	Carbon deposit on spark plug?	Clean or replace spark plug.
	Short circuit due to deficient spark plug insulation?	Check spark plug insulation, replace if worn.
	Improper spark plug gap?	Set to proper gap.
	Spark plug is red?	Check transistor ignition unit.
	Spark plug is bluish white?	If insufficient compression, repair or replace engine. If injected air leaking, correct leak. If carburetor jets clogged, clean carburetor.
	No spark present at tip of spark plug?	Check if transistor ignition unit is broken, and replace defective unit. Check if voltage cord cracked or broken and replace. Check if spark plug is fouled and replace.
	No oil?	Add oil as required.
	Oil pressure alarm lamp blinks upon starting? (if applicable)	Check automatic shutdown circuit, oil sensor. (if applicable)
Difficult to start, fuel is available, and spark is present at the spark plug.	ON/OFF switch is shorted?	Check switch wiring, replace switch.
	Ignition coil defective?	Replace ignition coil.
	Improper spark gap, points dirty?	Set correct spark gap and clean points.
	Condenser insulation worn or short circuiting?	Replace condenser.
	Spark plug wire broken or short circuiting?	Replace defective spark plug wiring.
Difficult to start, fuel is available, spark is present and compression is normal.	Wrong fuel type?	Flush fuel system, replace with correct type of fuel.
	Water or dust in fuel system?	Flush fuel system.
	Air cleaner dirty?	Clean or replace air cleaner.
	Choke open?	Close choke.
Difficult to start, fuel is available, spark is present and compression is low.	Suction/exhaust valve stuck or protruded?	Reseat valves.
	Piston ring and/or cylinder worn?	Replace piston rings and/or piston.
	Cylinder head and/or spark plug not tightened properly?	Torque cylinder head bolts and spark plug.
	Head gasket and/or spark plug gasket damaged?	Replace head and spark plug gaskets.
No fuel present at carburetor.	No fuel in fuel tank?	Fill with correct type of fuel.
	Fuel cock does not open properly?	Apply lubricant to loosen fuel cock lever, replace if necessary.
	Fuel filter/lines clogged?	Replace fuel filter.
	Fuel tank cap breather hole clogged?	Clean or replace fuel tank cap.
	Air in fuel line?	Bleed fuel line.



NOTES:

1. GROUND TO STOP ENGINE, (OIL SWITCH).
2. PINS M AND G GROUNDED TO IGNITION CASE.
3. AUDIBLE ALARM ORDER AS OPTION.
4. LOCATED INSIDE STEERING COLUMN.
5. LOCATED UNDERNEATH FOOT PLATFORM.
6. LOCATED IN ENGINE COMPARTMENT.
7. GROUND TO START ENGINE, (IGNITION SWITCH).

OPERATION MANUAL

HERE'S HOW TO GET HELP

PLEASE HAVE THE MODEL AND SERIAL NUMBER ON-HAND WHEN CALLING

UNITED STATES

Multiquip Inc.

(310) 537- 3700
6141 Katella Avenue Suite 200
Cypress, CA 90630
E-MAIL: mq@multiquip.com
WEBSITE: www.multiquip.com

CANADA

Multiquip

(450) 625-2244
4110 Industriel Boul.
Laval, Quebec, Canada H7L 6V3
E-MAIL: infocanada@multiquip.com

UNITED KINGDOM

Multiquip (UK) Limited Head Office

0161 339 2223
Unit 2, Northpoint Industrial Estate,
Globe Lane,
Dukinfield, Cheshire SK16 4UJ
E-MAIL: sales@multiquip.co.uk

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